

PS26162 — Updated Technical Documentation (SIH 2026, NTRO) - Reflecting Four-Class Model with Unclassified Fallback

Source material used: VIIRS_EDA_Findings_Report.docx [EDA report], Jupyter Notebook – generated with runcell.pdf [runcell PDF], osi-wri-data.ipynb [OSM/WRI nb], state_evidence_table.csv [CSV], PS26162_evidence_based_geographic_partition_selection.ipynb [partition nb]. Every number below is traceable to one of these files; items with no traceable value are marked **NOT FOUND IN SOURCE MATERIAL**.

Locked project decisions (stated as given, not re-derived)

- **Taxonomy Change (Revised):** The target system will be a four-class classifier for industrial, mining, agricultural_burn, and wildfire. The term "unclassified" is no longer a trained class. Instead, it is a post-training fallback mechanism where any detection whose top-class probability falls below a yet-to-be-validated confidence threshold will be routed for human review. This decision was made because the "unclassified" category lacks a coherent feature signature and training on it would degrade the model's ability to separate the four distinct fire types that do have real signatures .
- **Gas flare rejected as its own class** and folded into "industrial": even at the most permissive threshold tested, only 5–12 cells nationwide were confirmed near a real WRI gas/oil plant, far below a usable training threshold. The underlying features (saturation flag, persistence days, distance to WRI gas/oil plants) are retained as model inputs, not as a separate output class.
- **Geographic split:** TRAIN = Maharashtra, Karnataka, Madhya Pradesh, Punjab, Andhra Pradesh, Telangana. TEST A (deployment-realistic) = Gujarat, Tamil Nadu. TEST B (adversarial/generalization) = Jharkhand, Rajasthan. Chosen from the

corrected per-state evidence table and passed the hard minimum-support check (train $\geq$ 100 cells per class, test $\geq$ 30 cells per class, per class).

- **Model:** CatBoost (chosen over XGBoost for native high-cardinality categorical handling of H3 cells; Ordered Boosting's resistance to spatial leakage).
 - **Data window:** extending from the roughly 2-year window in the uploaded EDA (2024-08-01 $\rightarrow$ 2026-08-01) to approximately 4 years, bounded by both VIIRS sensors (S-NPP and NOAA-20) being simultaneously operational, balancing more training data against OSM-snapshot staleness risk and compute limits. This re-run has not happened yet; the uploaded files reflect the 2-year window only.
-

SECTION 1 — Technical Findings (ML/training pipeline)

1.1 Raw FIRMS VIIRS schema

Four source files :

Source	File	Satellite sample	acq_time dtype	type column
J1 archive	fire_archive_J1V-C2_794139.csv	N20	int64	Has type
J1 NRT	fire_nrt_J1V-C2_794139.csv	N20	int64	No type
SNPP archive	fire_archive_SV-C2_794140.csv	SNPP	int64	Has type
SNPP NRT	fire_nrt_SV-C2_794140.csv	SNPP	int64	No type

Raw archive column list : latitude, longitude, brightness, scan, track, acq_date, acq_time, satellite, instrument, confidence, version, bright_t31, frp, daynight, type; NRT identical minus type. acq_time validated as int64 HHMM (asserts: 0–2359, minutes < 60) .

Harmonization renames : brightness $\rightarrow$ bright_ti4, bright_t31 $\rightarrow$ bright_ti5, type $\rightarrow$ fire_type; confidence l/n/h $\rightarrow$ low/nominal/high; daynight D/N $\rightarrow$ Day/Night; satellite N20 $\rightarrow$ NOAA20, N $\rightarrow$ SNPP. Harmonized columns: latitude, longitude, bright_ti4, scan, track, acq_date, acq_time, satellite, instrument, confidence, version, bright_ti5, frp, daynight, fire_type, source, satellite_name, is_static_land, is_offshore. Derived labels: is_static_land = (fire_type==2),

is_offshore = (fire_type==3), both set to -1 where fire_type is NaN (all NRT rows) .

1.2 Ingestion & cleaning

- Total raw rows (archive + NRT, before overlap removal): **2,597,596** .
- Raw satellite split: N20 **1,329,378**; SNPP **1,268,218** .
- Exact duplicate rows: **0** .
- Archive/NRT overlap rows dropped (dedup key: latitude, longitude, acq_date, acq_time, satellite): **2** .
- Physical sanity filters (bright_ti4 > 200, frp >= 0) rows dropped: **0** .
- Rows after dedup + cleaning: **2,597,594** .
- Satellite after harmonization: NOAA20 **1,329,377**; SNPP **1,268,217** .
- Date range: **2024-08-01 to 2026-08-01** .
- Raw harmonized memory usage: **1326.00 MB** .
- Missing values (%): fire_type **4.65**; all other columns **0.00** . fire_type NaN count = **120,735** .

1.3 Physical distributions (point level)

Percentile table (count / mean / std / min / 1% / 5% / 25% / 50% / 75% / 95% / 99% / max / skew / kurtosis) :

col	count	mean	std	min	1%	5%	25%	50%	75%	95%	99%	max	skew	kurtosis
bright_ti4	2,597,596	331.403556	16.937914	207.93	297.34	301.73	316.93	335.35	342.4	354.01	367	367	-0.488448	0.000000
bright_ti5	2,597,596	301.597048	9.746338	243.18	279.24	286.78	294.12	302.04	308.47	316.69	323.76	387.83	0.076026	0.000000
frp	2,597,596	6.001282	13.162558	0	0.43	0.75	1.9	3.52	6.23	16.15	46.6	1074.64	15.836358	4.000000
scan	2,597,596	0.452376	0.081331	0.32	0.32	0.35	0.39	0.43	0.5	0.61	0.73	0.8	1.173411	1.000000
track	2,597,596	0.469152	0.109133	0.36	0.36	0.36	0.38	0.43	0.55	0.69	0.76	0.78	0.926966	-0.000000

- Saturated I4 records (≥ 366.9 K; threshold $367.0\text{ K} - 0.1$ buffer): **78,309 (3.01%)** . By satellite: SUOMI-NPP **39,909**; NOAA-20 **38,400** .
- Confidence: nominal **79.63%**, low **17.83%**, high **2.54%** .
- Day/night: Day **70.61%**, Night **29.39%** .
- fire_type distribution: 0.0 $\rightarrow$ **2,174,507**; 2.0 $\rightarrow$ **298,418**; NaN $\rightarrow$ **120,735**; 3.0 $\rightarrow$ **3,912**; 1.0 $\rightarrow$ **24** .
- 98th-percentile FRP threshold: **28.73**; high-FRP detections: **51,967** .

1.4 H3 aggregation (resolution 8, daily)

- Unique H3 cells: **921,202**; unique H3-days: **1,718,002** .
- Aggregations per (h3_08, acq_date): frp_max, frp_mean, n_detections, ti4_max, is_saturated_max, scan_mean, track_mean, confidence_high_any (any=="high"), daynight ("Both" if mixed), satellite_nunique, is_static_land (max), is_offshore (max) .

Top 20 cells by total detections :

h3_08	total_detections	active_days	max_frp	mean_frp
883ca9c95bfffff	4392	577	14.79	3.212537
883c817b29fffff	3863	557	18.36	3.304526
883c85d9c3fffff	3727	553	52.27	5.409623
883ca04021fffff	3647	572	19.26	3.462155
883ca826e9fffff	3432	554	13.53	2.950626
8842d9d2e5fffff	3134	601	41.76	5.96198
883c8d8ae9fffff	2713	565	12.51	3.277167
883c83b5d3fffff	2666	545	20.52	4.178771
8842d9d2e7fffff	2446	596	47.93	7.298662
883c854a19fffff	2421	529	11.84	2.727868
883c8e8ee7fffff	2377	513	47.05	3.992121
883c817b2dfffff	2315	510	15.69	2.694139
883ca9cb07fffff	2292	548	11.07	2.784386
883ca9d9a7fffff	2260	543	11.81	2.965118
883c85d9cbfffff	2247	511	42.79	4.862038
883cab6ce5fffff	2201	512	8.53	1.964667
8860a1b883fffff	2097	503	11.43	2.577912
883c817b23fffff	2060	449	14.74	3.317137
883ca0402dfffff	2049	492	15.1	2.93554
883ca83661fffff	2006	527	14.29	3.03543

Top 20 by active days :

h3_08	total_detections	active_days	max_frp	mean_frp
8842d9d2e5fffff	3134	601	41.76	5.96198
8842d9d2e7fffff	2446	596	47.93	7.298662
883ca9c95bfffff	4392	577	14.79	3.212537
883ca04021fffff	3647	572	19.26	3.462155
883c8d8ae9fffff	2713	565	12.51	3.277167
883c817b29fffff	3863	557	18.36	3.304526
883ca826e9fffff	3432	554	13.53	2.950626
8842d9d2e3fffff	1454	553	56.34	6.522289
883c85d9c3fffff	3727	553	52.27	5.409623
883ca9cb07fffff	2292	548	11.07	2.784386
883c83b5d3fffff	2666	545	20.52	4.178771
883ca9d9a7fffff	2260	543	11.81	2.965118
883c8d8a85fffff	1873	534	12.85	2.776045
883c84a909fffff	1917	533	13.02	3.414846
883ca9c86dfffff	1974	532	16.82	3.935917
883ca9c9c5fffff	1794	530	15.71	2.859146
883c854a19fffff	2421	529	11.84	2.727868
883ca83661fffff	2006	527	14.29	3.03543
883ca8366bfffff	1431	526	19.52	3.17707
883c8d2ec5fffff	1967	525	13.37	2.499962

Top 20 by max FRP :

h3_08	total_detections	active_days	max_frp	mean_frp
883cc1598bffff	16	2	1074.64	146.832545
883cea6883ffff	3	1	906.67	313.956667
883cc15a8bffff	1	1	890.08	890.08
883cea4a8bffff	11	1	868.25	149.822727
883c8054cbffff	1	1	820.37	820.37
883cc34aa5ffff	4	2	810.83	142.473333
883cc34aadffff	3	1	810.83	285.943333
883cc072cbffff	10	3	806.57	146.941905
883cc18257ffff	4	2	801.17	255.9275
883cc34c67ffff	13	2	801.01	138.615714
883cea2e8dffff	6	1	751.5	193.255
883cc34f1dffff	8	1	717.23	295.82875
883cc34f19ffff	10	1	717.23	272.587
883cccd123ffff	10	2	713.98	136.683125
883cc14725ffff	8	1	713.7	167.58875
883cc56c99ffff	5	1	696.49	303.26
883cc56cd3ffff	5	2	696.49	312.483333
883cc08c95ffff	2	2	685.38	388.49
883cc199b3ffff	4	1	655.38	330.275
883cc18b59ffff	2	1	655.38	330.25

1.5 Rolling (historical) features — Phase 8

Definition : within each H3 cell sorted by `acq_date`, `shift(1)` is applied **before** rolling (today's observation cannot leak into today's history); then groupby-rolling on `acq_date` with `min_periods=1`. Created features:

- `frp_max_lag7` = max of previous-observation `frp_max` over trailing 7D window
- `frp_max_lag30` = same over 30D
- `active_days_7d / 30d / 90d` = sum of previous active-day indicators over trailing window (no `frp_max_lag90` by design)

Summary statistics :

feature	count	mean	std	min	50%	75%	90%	95%	99%	max
frp_max_lag7	796,800	6.144644	12.647896	0	3.74	6.54	11.16	16.42	44.3201	810.83
frp_max_lag30	796,800	6.690611	13.261621	0	4.11	7.13	12.06	17.88	47.81	810.83
active_days_7d	1,718,002	0.880859	1.490189	0	0	1	2	5	7	7
active_days_30d	1,718,002	2.185428	5.610345	0	0	1	4	17	28	30
active_days_90d	1,718,002	5.008328	15.38911	0	0	1	8	42	81	90

Missingness: `frp_max_lag7` and `frp_max_lag30` each NaN on exactly **921,202** rows (the first observed day of each H3 cell); active-day features have **0** NaN . Leakage validation: first-day check and chronological-order check passed ("PHASE 8 LEAKAGE CHECK PASSED") . Pipeline note: NaNs are kept intentionally for CatBoost native missing handling — do **not** `fillna(0)` (collides with real FRP=0) .

1.6 OSM context features [OSM/WRI nb]

- Source: Geofabrik `india-latest.osm.pbf`, **1.71 GB** at run time; filtered with `osmium tags-filter`.
- Tags used: `landuse=industrial`, `landuse=quarry`, `landuse=farmland`, `man_made=mineshaft`, `man_made=adit`, `power=plant`, `power=substation`, `power=generator`.
- Filtered features nationwide: **415,037**. Per category: industrial **54,521**; quarry **20,855**; farmland **213,931**; mineshaft **1,017**; adit **3**; power_infra (`plant+substation+generator`) **119,648**.
- Features created per category: `dist_osm_<cat>_km` (haversine distance to nearest category-feature centroid) and `n_osm_<cat>_5km` (count within 5 km).

1.7 WRI power-plant features [OSM/WRI nb]

- WRI India plants (all fuels): **1,589**. Breakdown: Solar **851**, Coal **253**, Hydro **233**, Wind **108**, Gas **68**, Biomass **50**, Oil **17**, Nuclear **9**.
- Features created per fuel: `dist_wri_<fuel>_km` (nearest plant of that fuel) and `n_wri_<fuel>_10km` (count within 10 km) — 16 columns total (solar, coal, wind, gas, hydro, biomass, oil, nuclear).

1.8 Final feature table (Phase 9) [EDA report][OSM/WRI nb]

- Rows: **1,718,002**; memory **421.78 MB** before downcasting, **183.83 MB** after.
- `is_first_observation` flags **921,202** rows; labeled H3-days **1,644,273** (**95.71%**); unlabeled (`is_static_land == -1`) **73,729** (**4.29%**).

- Target distribution: `is_static_land` 0 → 1,494,838; 1 → 149,435; -1 → 73,729.
- Feature manifest: identifiers `h3_08`, `acq_date`; target `is_static_land`; label mask `is_labeled`; categorical `daynight`; boolean `confidence_high_any`; numeric (15) `frp_max`, `frp_mean`, `n_detections`, `ti4_max`, `is_saturated_max`, `scan_mean`, `track_mean`, `satellite_nunique`, `is_offshore`, `frp_max_lag7`, `frp_max_lag30`, `active_days_7d`, `active_days_30d`, `active_days_90d`, `is_first_observation`.
- Final dtypes: `h3_08` category; `acq_date` `datetime64[ns]`; `frp_max`/`frp_mean`/`ti4_max`/`scan_mean`/`track_mean`/`frp_max_lag7`/`frp_max_lag30` `float32`; `n_detections` `int16`; `is_saturated_max`/`satellite_nunique`/`is_static_land`/`is_offshore`/`active_days_7d`/`30d`/`90d`/`is_first_observation`/`is_labeled` `int8`; `confidence_high_any` `bool`; `daynight` category.
- Saved: `sih2026_h3_daily_features.parquet` (round-trip verified). After merging state + OSM + WRI context: **shape (1,718,002 × 52)**, saved as `sih2026_h3_daily_features_with_osm_wri.parquet`. The 31 added columns are `h3_lat`, `h3_lon`, `state`, the 16 WRI columns (§1.7) and 12 OSM columns (§1.6).
- State assignment: 36 states/UTs loaded; **91** of 921,202 cells matched >1 state (resolved by nearest distance); **1,882 (0.20%)** cells unmatched (offshore/coastal); **35** states represented .
- Note for the pipeline builder: the executed artifacts carry only the binary `is_static_land` label (+ -1 unlabeled). An executed multi-class label column for the locked 5-class taxonomy is **NOT FOUND IN SOURCE MATERIAL**.

1.9 Corrected state-level evidence table [CSV]

Reproduced verbatim from `state_evidence_table.csv` (the corrected artifact; it matches the rendered heatmap. The notebook's inline printed table in cell 12 of [OSM/WRI nb] shows different, pre-correction counts and should not be used).

state	n_h3_days	n_unique_cells	industrial	quarry	farmland	power_infra	wri_solar	wri_coal	wri_wind	wri_gas	wri_hydro
Andaman & Nicobar	79	60	2	2	1	3	3	0	0	0	0
Andhra Pradesh	85890	54363	8881	3671	8508	13296	3556	304	265	443	974
Arunachal Pradesh	11139	7248	203	117	1442	769	0	0	0	6	111
Assam	31505	17564	2032	1419	2952	2136	0	35	0	84	421
Bihar	39268	18929	2019	786	1421	3516	45	240	0	0	2
Chandigarh	12	8	8	0	2	8	0	0	0	0	0
Chhattisgarh	184402	80329	5734	3388	5908	10116	1296	3887	0	0	109
Daman & Diu & Dadra & Nagar Haveli	117	86	69	7	34	68	2	0	0	0	0
Delhi	812	336	310	2	329	306	26	46	0	177	0
Goa	919	526	491	486	480	378	0	0	0	49	0
Gujarat	49150	27471	8368	1656	1991	18130	1719	440	497	786	202
Haryana	25400	16702	4920	552	2692	9687	841	441	0	73	30
Himachal Pradesh	8689	6140	996	418	1166	2175	8	109	0	0	1484
Jammu and Kashmir	8027	5351	298	148	288	1204	0	0	0	0	504
Jharkhand	72039	18695	3914	4555	964	2443	34	783	0	0	104
Karnataka	84865	52749	14511	5769	7745	29012	9195	1678	1490	72	748
Kerala	3633	2562	965	986	1959	1290	29	0	57	18	291
Ladakh	20	16	5	3	15	9	0	0	0	0	3
Madhya Pradesh	268613	142173	9666	11292	9374	25567	2300	3899	432	173	1207
Maharashtra	166883	105610	18424	10723	18713	28808	3660	3064	2328	813	3698
Manipur	21331	11858	417	39	3276	1302	0	0	0	0	244
Meghalaya	17043	10122	360	364	816	934	0	0	0	0	765
Mizoram	19228	11295	295	93	679	1219	0	0	0	0	141
Nagaland	9446	6186	185	181	622	701	0	0	0	0	114
Odisha	159847	72980	7293	3093	3151	10614	1512	1219	0	0	894
Puducherry	196	60	58	0	58	54	10	0	0	29	0
Punjab	79470	40300	18130	923	7980	26699	5301	1118	0	0	199

state	n_h3_days	n_unique_cells	industrial	quarry	farmland	power_infra	wri_solar	wri_coal	wri_wind	wri_gas	wri_hydro
Rajasthan	56445	27650	3571	2066	1009	7571	2016	1275	298	569	193
Sikkim	123	107	35	2	86	58	0	0	0	0	67
Tamil Nadu	28701	19912	8004	3092	5431	10438	2208	259	527	167	494
Telangana	96648	54794	8224	2442	6348	18739	9832	677	0	0	903
Tripura	6970	3998	234	139	94	287	10	0	0	202	95
Uttar Pradesh	112264	72308	14079	1963	7945	14412	2058	1558	0	315	377
Uttarakhand	23220	15334	2153	776	2630	3313	697	0	0	76	1119
West Bengal	41606	15498	2236	549	1609	3399	678	652	0	0	210

The hard-support validator `check_partition_support(train, testA, testB, ...)` with `min_train=100`, `min_test=30` per class exists in [OSM/WRI nb] and is the check the locked split passed.

1.10 Gas-flare evaluation (threshold sensitivity)

Executed results at the only cutoff tabulated in the uploaded outputs (`is_saturated_max == 1` AND `active_days_90d >= 60`) :

- Raw candidates (saturation + persistence): **443 H3-days, 41 unique cells**.
- WRI-confirmed (within 5 km of a WRI gas/oil plant): **5 unique cells** (all in Gujarat per §1.9).
- Unexplained (saturated + persistent, no nearby WRI plant): **36 unique cells**.
- `active_days_90d` distribution among ALL 61,610 saturated H3-day rows: mean **1.19565**, std **7.11881**, min **0**, 25% **0**, 50% **0**, 75% **1**, max **90** .
- Per-state candidate/confirmed counts at this cutoff: see `flare_cand` / `flare_conf` columns in §1.9.

NOT FOUND IN SOURCE MATERIAL: raw-vs-confirmed counts at other `active_days_90d` cutoffs. Only the ≥ 60 cutoff was executed in the uploaded outputs; the "5–12 confirmed at the most permissive threshold" figure comes from the locked decision record, stated as given in the preamble.

1.10a Wildfire / agricultural_burn label-ordering issue (identified, unresolved)

The current Phase 6 bootstrap label rule checks farmland proximity before wildfire intensity. This means a genuinely intense wildfire detection occurring near a farm field

will be labeled `agricultural_burn` purely due to rule execution order, not because it is actually a farm burn. A proposed fix — checking FRP intensity and detection suddenness BEFORE farmland proximity, so genuinely intense fires win the tiebreak even near farmland — has been discussed but NOT implemented and NOT validated against real data. Validating it requires a groupby comparison of `frp_max` for near-farmland vs. not-near-farmland cells to find a real, data-derived intensity threshold, rather than an assumed one. This is an open item, not a locked decision.

1.11 Source-material gaps register

- [partition nb] was uploaded **unexecuted** (no cell outputs). None of its artifacts (empirical spatial buffer km, correlogram, WorldClim/WorldCover/ecoregion descriptors, top-5 partitions from that pipeline, `final_geographic_split.json`) exist as executed numbers. The locked geographic split stands as given (§preamble).
 - Executed 5-class label table: **NOT FOUND** (see §1.8 note).
 - Flare sensitivity series beyond the ≥ 60 cutoff: **NOT FOUND** (see §1.10).
 - Confidence-threshold cutoff for the unclassified fallback: not validated against a real model.
 - `power_plant` as a possible split from industrial: proposed, requires the same kind of per-state WRI density support-check that `gas_flare` received, never executed.
 - Data window extension from ~ 2 years to ~ 4 years: decided in principle, **NOT yet re-executed**. All numbers in this document still reflect the original 2024-08-01 to 2026-08-01 window only.
 - VIIRS processing-version consistency check across a wider time window: flagged as needed before trusting an extended dataset, not yet run.
 - Ecological/spatial-leakage-aware partition validation (the separate `evidence_based_geographic_partition_selection` notebook): still unexecuted. Current state split passed only the simpler hard-support-count check, not this more rigorous method.
 - An executed multi-class (4-class) label column: does not exist yet against either the 2-year or planned 4-year dataset. Phase 6 labeling has not been run for real.
-

SECTION 2 — Practical Summary (for backend / Sagar)

2.1 Raw inputs and refresh cadence

The pipeline batch-ingests four external sources: (1) NASA FIRMS VIIRS thermal-anomaly CSVs — archive + near-real-time (NRT) files for both satellites (NOAA-20/J1 and Suomi-NPP), ~2.6M rows for the executed 2024-08-01 → 2026-08-01 window; (2) OpenStreetMap India extract (`india-latest.osm.pbf`, 1.71 GB at run time) filtered to industrial/quarry/farmland/mine/power features; (3) WRI Global Power Plant Database (CSV, 1,589 India plants); (4) an India state/UT boundary shapefile (36 units). **No scheduled pull cadence is specified in the source material** — the uploaded pipeline is a full-window batch re-run; a ~4-year window re-run is planned but has not happened yet, so current artifacts cover the 2-year window only.

2.2 The final per-H3-cell-day table (52 columns)

One row per H3 cell (~0.7 km² hexagon) per day in which a thermal detection occurred. Columns, in plain language:

Column(s)	Meaning
<code>h3_08, acq_date</code>	The hexagon ID and the date of the detection(s).
<code>h3_lat, h3_lon, state</code>	Hexagon centre coordinates and the Indian state/UT it falls in.
<code>frp_max, frp_mean, n_detections</code>	Strongest fire-radiative-power reading that day, the average, and how many detections fell in the cell.
<code>ti4_max, is_saturated_max</code>	Hottest brightness temperature recorded; flag for whether the sensor reading maxed out (saturated).
<code>scan_mean, track_mean</code>	Average sensor view geometry (quality proxies, not locations of anything on the ground).
<code>confidence_high_any, daynight, satellite_nunique</code>	Whether any detection was high-confidence; day/night pass; whether one or both satellites saw it.
<code>frp_max_lag7/30, active_days_7d/30d/90d</code>	History of this same cell: hottest reading in the previous 7/30 days and how many of the past 7/30/90 days it was active (persistence).
<code>is_first_observation, is_labeled, is_static_land, is_offshore</code>	Housekeeping/label columns: first-ever sighting of the cell; whether the archive provides a label; archive flag for static industrial-type source; offshore flag.
<code>dist_osm..._km (6)</code>	Distance from the cell to the nearest mapped feature of each type (industrial, quarry, farmland, mineshaft, adit, power_infra).
<code>n_osm..._5km (6)</code>	Count of mapped features of each type within 5 km.
<code>dist_wri..._km (8)</code>	Distance to the nearest power plant of each fuel type (solar, coal, wind, gas, hydro, biomass, oil, nuclear).
<code>n_wri..._10km (8)</code>	Count of power plants of each fuel type within 10 km.

2.3 Classification output

Each row will receive one of the four trained class labels: **industrial**, **mining**, **agricultural_burn**, **wildfire**, plus a confidence score (per-class probabilities from the CatBoost model). Any detection where the model's top-class probability falls below a yet-to-be-validated confidence threshold will be routed to the **unclassified** fallback bucket for human review.

Classification output schema: NOT FOUND IN SOURCE MATERIAL. While the design intent is to provide per-class probabilities and a clear routing reason, the specific field names and structure for this output have not been finalized and executed.

2.4 Data-quality caveats backend should know

- **OSM and WRI are present-day snapshots** joined onto two years of historical detections; features that change over time (new mines, plant closures) are not dated. This staleness risk is acknowledged in the locked 4-year-window decision.
- **NRT rows carry no archive-only type field**, so 73,729 H3-days (4.29%) are unlabeled (`is_static_land = -1`); raw `fire_type` missingness is 4.65%.
- **1,882 H3 cells (0.20%)** could not be assigned to any state (offshore/coastal), and 91 cells matched two states and were resolved by nearest distance; expect a small number of rows with missing/derived state.
- **Gas flare is not a separate output class** (locked): only 5 cells nationwide were confirmed near real gas/oil plants at the tested threshold; flare-related columns remain as inputs only.
- Lag columns (`frp_max_lag7/30`) are legitimately empty on a cell's first sighting and flagged via `is_first_observation`; downstream consumers must not treat these as errors or zero-fill them.
- WRI gas/oil coverage in India is thin (68 gas + 17 oil plants), which is why flare confirmation counts are single-digit in some states (e.g., Gujarat is the only state with confirmed cells: 5).
- The **unclassified** fallback routing mechanism is a conceptual design intent, not an implemented feature. The specific confidence threshold value required to trigger this fallback has **NOT BEEN VALIDATED** and is therefore unknown.

Reference

1. SIH 2026 Problem Statements Analysis <https://www.kaggle.com/code/devraai/sih-2026-problem-statements-analysis/input>
2. Smart India Hackathon 2026 problem statements release ... <https://www.facebook.com/rcciitofficial/posts/problem-statements-for-the-9th-edition-of-the-smart-india-hackathon-sih-2026-wil/1677964841001998/>
3. SIH 2026 — Need advice from previous winners/finalists ... https://www.reddit.com/r/hackathon/comments/1vjx1sg/sih_2026_need_advice_from_previous/
4. SIH 2026 Problem Statements Explorer <https://www.codehuntersacademy.com/sih-2026-ps>
5. SIH 2026 Problem Statements Overview | PDF | Agriculture <https://www.scribd.com/document/913856423/Problem-Statements-Smart-India-Hackathon-Sih>
6. SIH 2026 Problem Statement Finder Tool | Vedant Chalke ... https://www.linkedin.com/posts/vedantchalke36_github-vedantchalke36sih-2026-problem-statements-activity-7496591710144241665-lZM2
7. Wildfire Suppression Difficulty Index 97th Percentile (2026) <https://data-nifc.opendata.arcgis.com/datasets/17104b8eab1e48ccb442414e6a6ef576/about>
8. SIH 2026 problem statements - GitHub <https://github.com/vedantchalke36/sih-2026-problem-statements>
9. ISO 3941:2026 - Fire Classification (Updated Standard ... - Facebook <https://www.facebook.com/Fireengineering/posts/iso-39412026-fire-classification-updated-standard-2026-editionsunderstanding-fir/782975574865505/>
10. SIH 2026 Problem Statements - Browse All 226 <https://sih2026.vuce.in/>
11. SIH 2026 Problem Statements Analysis | Kaggle <https://www.kaggle.com/code/devraai/sih-2026-problem-statements-analysis>
12. Wildfire Suppression Difficulty Index 97th Percentile 2026 (Image Service) https://agdatacommons.nal.usda.gov/articles/dataset/Wildfire_Suppression_Difficulty_Index_97th_Percentile_2024_Image_Service_/25972480
13. Top 5 Industrial Fire Safety Solutions to Implement in 2026 | India Guide <https://cholarisk.com/blog/top-5-industrial-fire-safety-solutions-to-implement/>

14. ISO 3941:2026 Fire Classification Update: Class L Lithium-Ion Fires https://www.linkedin.com/posts/akash1997_iso-39412026-fire-classification-updated-activity-7431218446849257472-eql
15. Wildfire Suppression Difficulty Index 97th Percentile 2026 (Image Service) <https://catalog.data.gov/dataset/wildfire-suppression-difficulty-index-97th-percentile-2025-image-service>
16. Fire Data Sets - USFA.FEMA.gov <https://www.usfa.fema.gov/statistics/data-sets/>
17. Smart India Hackathon 2026 — All 226 Problem Statements ... [https://www.blinknbuild.in/Assets/](https://www.blinknbuild.in/Assets/SIH_2026_All_226_Problem_Statements_Master_Catalogue.pdf)
[SIH_2026_All_226_Problem_Statements_Master_Catalogue.pdf](https://www.blinknbuild.in/Assets/SIH_2026_All_226_Problem_Statements_Master_Catalogue.pdf)
18. SIH 2026 Problem Statements Dataset <https://www.kaggle.com/datasets/riteshmaurya86/sih-2026-problem-statements-dataset>
19. SIH — Your Gateway to Smart India Hackathon <https://amityhackathon.vercel.app/problems>
20. Smart India Hackathon (SIH) 2026 Problem Statements in ... <https://github.com/NoBugNinja/Smart-India-Hackathon-SIH-2026-Problem-Statements>
21. SIH Buddy: SIH 2026 Problem Statements — All 226, Scored <https://sih-buddy.vercel.app/>
22. Affordable Wearable Medical Devices | PDF <https://www.scribd.com/document/986055420/TN-IMPACT-Problem-statement>
23. dmdb › chandra › Enron2.1 › words <https://www.ics.uci.edu/~dmdb/chandra/Enron2.1/words.txt>
24. Official Problem Bank - Amity Internal Hackathon <https://amityhackathon.vercel.app/problems?theme=miscellaneous>
25. Dramatic Compositions, Motion Pictures 1934: Vol 7 No 1-12 https://archive.org/stream/catalogofcopyrig71libr/catalogofcopyrig71libr_djvu.txt
26. Chatbot Arena Query Topic Modelling - Discover gists - GitHub <https://gist.github.com/shekharx/e1c36ed62b119fd30e739ee8bc589468>
27. SIH 2026 Problem Statement Analysis | Do NOT Pick One Until You ... <https://www.youtube.com/watch?v=1pGlohRbBnc>
28. USAR SIH 2026 Internal Management Portal - Apps Script https://script.google.com/macros/s/AKfycbzhitza8k3P9F0mUaj_7v7EaXYL1q0txXkppBTQU1aYIPFUtvjW2HhJzxL1SP_obgwYwg/exec
29. FSQs for (SIH) 2026 – FREQUENTLY ASKED QUESTIONS (FAQs) Still ... <https://www.facebook.com/karthik.n.iyer/videos/-fsqs-for-sih-2026-frequently-asked-questions-faqs-still-confused-about-sih-2026/4490651374587995/>

30. [XLS] Worksheet - Smart India Hackathon <https://www.sih.gov.in/letters/problemStatements.xlsx>
31. SIH 2026 Problem Statements : How & Where to Analyze | Choose With AI https://www.youtube.com/watch?v=dIPHrD0p_uE
32. When will SIH 2026 Problem Statements be released? - Reddit https://www.reddit.com/r/Indian_Academia/comments/1vpq494/when_will_sih_2026_problem_statements_be_released/
33. Smart India Hackathon 2026 - GGSIPU EDC <https://sites.google.com/view/ggsipuedc/students/student-council/events/sih-2026>
34. ISO 3941:2026 Fire type classification update | Ergasia PRL <https://www.ergasia.es/iso-39412026-fire-type-classification-update-n-63-en.html>
35. ISO 3941:2026 Fire Classification Standard Update - LinkedIn https://www.linkedin.com/posts/dhilephan-m_so-39412026-updated-fire-classification-activity-7427336428684038145-TOLu
36. ISO 3941:2026(en), Classification of fires <https://www.iso.org/obp/ui#!iso:std:iso:3941:ed-3:v1:en>
37. Serious Injury and Fatality – ASTM activity for 2026–2027 – IHMM <https://ihmm.org/serious-injury-and-fatality-astm-activity-for-2026-2027/>
38. Fire Classes & Extinguisher Selection Reference 2026 - SmartQHSE <https://www.smartqhse.com/datasets/fire-classes-extinguishers-2026>
39. Iso 3941 2026 | PDF | International Organization For Standardization <https://www.scribd.com/document/993621718/iso-3941-2026>
40. Essential 2026 Fire Safety Code Updates: What Every Affiliate ... <https://blog.commercialfire.com/blog/essential-2026-fire-safety-code-updates-what-every-affiliate-needs-to-know>
41. ISO 3941:2026 - Fire Classification (Updated Standard ... - Instagram <https://www.instagram.com/p/DUkE71nidbI/?hl=en>
42. The Ultimate Guide to Industrial Fire Safety in 2026 <https://maxserveglobal.com/blog/the-ultimate-guide-to-industrial-fire-safety-in-2026/>
43. Smart India Hackathon <http://www.sih.gov.in/>
44. SIH 2026 Series | Part 1 If you're a college student, Smart India ... <https://www.instagram.com/reel/DbiJfjTvhr2/>
45. Curated all SIH 2026 problem statements into a searchable format https://www.reddit.com/r/Btechtards/comments/1vvgb30/curated_all_sih_2026_problem_statements_into_a/
46. SIH 2026 Problem Statements Are Announced! The wait is over! Smart ... <https://www.instagram.com/p/DcUq4TCBA0h/>

47. Fire Classifications | Class A, B, C, D, and K Fire <https://hct-world.com/fire-classifications/>
48. Fire Classes in India: Class A, B, C, D, E, F Explained with Extinguisher ... <https://jsnmengineers.in/blog/fire-classes-india-types-explained>
49. What Are the 5 Different Classes of Fire — Pye-Barker Fire & Safety <https://pyebarkerfs.com/what-are-the-5-different-classes-of-fires/>
50. Fire Classification as per IS 15683 : 2018 (BIS) Indian Standard ... <https://www.instagram.com/p/DSuPF7lDLvk/?hl=en>
51. [XML] CODE OF PRACTICE FOR FIRE SAFETY OF BUILDING (GENERAL) <https://law.resource.org/pub/in/bis/S03/is.1641.1988.html>
52. Classes of Fires | Class A, B, C, D, and F Fire - Kanex Fire <https://www.kanexfire.com/blog/know-about-the-classes-of-fire/?srsltid=AfmBOoo3qJsVJG4Ns5qjel7lO39s4rJtpZwkwhwcwWGGlms5tesMwxcu>
53. Protection Strategies for Industrial Facilities in India - Safex Fire Services <https://safexfire.com/d-fire/>
54. Classification of Fire as per Indian Standard - NFSC Study Desk <https://www.firestudy.in/2024/12/classification-of-fire-as-per-indian.html>
55. The 6 Types And Classes Of Fire (And How To Put Them Out) <https://www.haspod.com/blog/fire/classes-of-fire>
56. ISO 3941:2026 - Fire Classification by Fuel Type for ... https://standards.iteh.ai/catalog/standards/iso/b09c882d-0fea-4fa0-ae37-610faf4e11e9/iso-3941-2026?srsltid=AfmBOoqjFbIGj37HUzmpiFWfwgrh_CHNgb10B8zKsGPLn7Fb2bwbyL9
57. 2026 Fire Protection Industry Trends & Insights <https://www.inspectpoint.com/2026-fire-life-safety-industry-report-key-trends-shaping-fire-protection/>
58. ISO 3941:2026 Fire Classification Standard | Yazid Bin ... https://www.linkedin.com/posts/yazid-bin-ismail_iso-39412026-classification-of-fires-this-activity-7422340221666521089-EDOO
59. The List of 300+ Codes and Standards <https://www.nfpa.org/for-professionals/codes-and-standards/list-of-codes-and-standards>
60. Class L Fires: What the New ISO 3941:2026 Classification ... <https://eticaag.com/class-l-fire-classification-iso-3941-lithium-ion-bess/>
61. What is a Class L Fire? Understanding ISO 3941:2026 and ... <https://hct-world.com/what-is-a-class-l-fire-understanding-iso-3941-2026-and-lithium-ion-battery-hazards/>
62. Discovery/DNS/bitquark-subdomains-top100000.txt <https://gitlab.com/kalilinux/packages/seclists/blob/f8146198d2359eaaabd0bd18b4afa880965449e8/Discovery/DNS/bitquark-subdomains-top100000.txt>

63. Text-Emotion-Analysis/MLModels.ipynb at master <https://github.com/kanchitank/Text-Emotion-Analysis/blob/master/MLModels.ipynb>
64. First-time SIH 2026 participant — how does evaluation work & how early ... https://www.reddit.com/r/hackathon/comments/1t0lk9d/firsttime_sih_2026_participant_how_does/
65. How to Crack SIH 2026 Internal Hackathon | Build a ... - YouTube <https://www.youtube.com/watch?v=721Tj06jIKw>
66. ISO 3941:2026 - Fire Classification by Fuel Type for Firefighting https://standards.iteh.ai/catalog/standards/iso/b09c882d-0fea-4fa0-ae37-610faf4e11e9/iso-3941-2026?srsId=AfmBOorDLOQbdi7U6oahR3CLwmOkRYwznxMRSJK2Oyukjo_gVW0E6JzF
67. OSHA Fire Safety Regulations 2026: What Every Employer Must Know <https://fireandsafetyjournalamericas.com/osha-fire-safety-regulations-2026-guide/>
68. Classes of Fire Chart Explained – A Fire Safety Guide for Homes and ... <https://swartzfireandsafety.com/posts/classes-of-fire-chart-explained-a-fire-safety-guide-for-homes-and-businesses>
69. Observance of Fire Service Week 2026 Fire Service Week ... - Facebook
CAESzQEB6zswFVwp_M4Q9ZjHoitg2aq7yCgp02ry61SDBLsNf7Xx9a8zx5EO1zeQUoW-WA-ATHiSnGRtx-TMAh5iyf5EqhFJfAd08ulD3aOBDhntd9reELQN-QFiITATY5D214MxeXbnkgDDsvCUCRBZ-0gnX12xiqgLPmUZ9j8YRZVQ33yZHSYvKiQLIhzcgkL35lEJHA5fyLSXQmRXpIWt-TgFU4wWq2sKJKsgOt10ItUkkqypnUS6uDV6SclOVrGfy6_gnrYokM9veYBTpA-5
70. National Fire Service Week 2026 (April 14–20) focuses on “Safe ...
CAESbQHrOzAVM2izwIkLSyqHq-mkG1t4s1V4_G55g8TnwaiaT1t7DQl4XvrTbMMLeS-WnfXEtLJlcyvqf2ujbLxKpwR1s7vMR5xYx_nAckhJuyKK4nE_tyfC3EvM8Y4bO3tlPW2fjO9zKejXcGfvFU
71. Fire India 2026 | Fire Safety, Rescue & Emergency Response Expo
CAEScQHrOzAVmE5er4B2XV-Rskh9KKjFO2wkVAC9Xk5Z5BqhpEx0Juf7AJudYQV1QYTn3AE2ojwuxJR_v7KP3AtqZtEk6TvEUJjqnMA3-F0lG1eQuk46w6USgJ8fbgkrGKI1bl4YFSPcRR5lEsQoK3aWTWuy
72. SIH 2026 Problem Statements Explorer - CodeHunters Academy
CAESZgHrOzAVvxvMFWRXEANf_oLRpITpgv4RxldKqszBiNWknmD7CFq60IAzEIVV0e-nEh585Ivp3xWI_1XfQuh7uHNTX_OcALWr0tVMClEO8aUHxUWmvtECBdE4fZ17mpsPBnl2S3cOag
73. Industrial Fire Summit - FDIC International 2026 CAESYwHrOzAVzyf-HAoT8z1etEhdsvp4EYqA_wJTM6858uwfapag4W1zUb8St4IByhM5aG1FNW8_o7OxFxCFWzx8L_cgVH7OJaihRP6yqH3a_tmIGLka9cFd9ks_0euqX_mw9EbaA
74. National Fire Service Week 2026 | Theme, Causes & Prevention Tips
CAESZwHrOzAVSXrYCWN-gGyw-

lkPbL3jJ9TRPQ-7gvJG0jwk6iIs_ukV5lPPXKip8LJHho4a8pWlD4OLwYZGucESkUkA5Ddl
fK_fxjefShT6s9MKjyBKup0YT_GhCeGavILSH8B3Gv9mfAM

75. Sih Live - Instagram

CAESYwHrOzAVHq5sNwUAhqXYIAUpJYVBFtWlEztF1o4ucZ4io2FALu0NFid1ZxU76AT
crO7I7EtvFnUDDlRQZTs434asKSeIXbrv-ijJNe1hgYJhJHIQS5xouQNeFHe6pIOD_INTSg

76. SIH 2026 Registration Process Explained! Confused about how to ...

CAESYAHrOzAVrHqZF6GHDsGqzjDfHxr2MejbbS3o6VtaSevys61g8TF4Ry8N7_Cso7u3
Tf9ksWlvAkSgr0w-jHYejNZDfhwf2Ge_3pob_M9XK91lFq4LDzGm23_8bD9ixy8Dvw

77. BASIC INDUSTRIAL FIRE FIGHTING TRAINING | 27-28 AUGUST 2026 ...

CAESxgEB6zswFSBLOctDMEzQyfCUAJUrt9T08TCXTCEBDzSI6x-
RLavvh23qZjNqumJCQ25ZakhcNHL4lc_ArR0GXRwcTeJZO73l3oOyLJdMYXeFTnzraaR
kMWq8x4f3nBFyJ-1QYKxgnXTnXq2rPFMgdG5-GxF78_sRB5PjhqrZZ4-
Bcz9nNeMp2aNVzZyFubljuKPNLfA3oO9ASA_LqjjJQSP14ZPqWy1Rocc00YGSU6IA91x
A7nMLJYgSKFDB6FRInQj3BRBTsD4

78. Proud Achievement Alert! We're thrilled to share that NEFSA has ...

CAESxQEB6zswFQazvVwP-
iwYimbqzabwdzEVasSXIPro4hX1nUJHJKe_F6o5_ulPJThQr5Em5JrnmSLW171aY_9yn
TksuX_STzbAtoUecS48qJgc_hmv1_yLlqXmE4I70B7ErgQ6z1tKcbkWWM5_OUO_P1mP
ofJW8bU-lrkormkoYwo9jhw7YxFLoDqcoLFtJZ2wzAtdjoUwV5TRF-
lCk3KgLOQaJGDbBZGx425kfNqE5F_1d79pYBv6PISJYg5qvNt_LInIiIXi_Q

79. sih · GitHub Topics <https://github.com/topics/sih?l=javascript&o=desc&s=updated>

80. government-of-india · GitHub Topics <https://github.com/topics/government-of-india>

81. SIH 2026 — Software Problem Statements - GitHub <https://github.com/Rugved-dev18/SIH-2026-official-Software-Problem-Statements/blob/main/README.md>

82. picoCTF 2026 / Cylab - Forensics Git 1 - Filesystem Extraction - YouTube <https://www.youtube.com/watch?v=tc3ixHJ9Sbs>

83. Deprecations and removals by version - GitLab Docs <https://docs.gitlab.com/update/deprecations/>

84. City of Sioux Falls - Metrics - Fire Incidents - Catalog - Data.gov <https://catalog.data.gov/dataset/fire-incidents>

85. INFORMATION.SCHEMA.COLUMNS - InterSystems IRIS Data Platform ... <https://docs.intersystems.com/irislatest/csp/documatic/%25CSP.Documatic.cls?LIBRARY=%25SYS&CLASSNAME=INFORMATION.SCHEMA.COLUMNS>

86. SIH 2026 problem statements are out! Before you lock one in ... - Instagram https://www.instagram.com/reel/DcXx6g1v_Ww/

87. SIH 2026: Understand Problem Statements Before Innovating https://www.linkedin.com/posts/tejaswinihbbat_sih2026-smartindiahackathon-sih-activity-7496865806610493440-UD--

88. Shortlisted teams of SIH 2025 <https://www.sih.gov.in/sih2025/screeningresult-batch1>
89. Shortlisted teams of SIH 2024 <https://sih.gov.in/sih2024/screeningresult>
90. misc_ml_codes/w2v_word2idx.json at master https://github.com/dhararkajit/misc_ml_codes/blob/master/w2v_word2idx.json
91. Runcell Features - | AI-Powered Jupyter Notebook Enhancement <https://www.runcell.dev/features>
92. Hybrid ML Pipeline for Log Analysis | PDF - Scribd <https://www.scribd.com/document/883322939/Hybrid-ML-Pipeline-Workflow>
93. Day 144: Building Production ML Pipelines for Log Intelligence <https://sdcourse.substack.com/p/day-144-building-production-ml-pipelines>
94. Production Machine Learning Pipeline for Text Classification with fastText <https://valohai.com/blog/production-machine-learning-pipeline-text-classification-fasttext/>
95. Videos - RunCell <https://www.runcell.dev/videos>
96. A Comprehensive Review and Evaluation of Classification Models https://www.researchgate.net/publication/376839986_Machine_Learning-Based_Fire_Detection_A_Comprehensive_Review_and_Evaluation_of_Classification_Models
97. Machine Learning-Based Fire Detection: A Comprehensive ... - JOIV <https://joiv.org/index.php/joiv/article/view/2332>
98. ML Pipelines - Spark 4.2.0 Documentation <https://spark.apache.org/docs/latest/ml-pipeline.html>
99. A Deep-Learning-Based Approach to the Classification of Fire Types - MDPI <https://www.mdpi.com/2076-3417/14/17/7862>
100. [PDF] Pipe Leakage Detection Use Case - arXiv <https://arxiv.org/pdf/2205.02706>
101. Avoiding Jupyter Notebooks entirely and doing everything ... https://www.reddit.com/r/datascience/comments/1ajw9s8/avoiding_jupyter_notebooks_entirely_and_doing/
102. Run | Databricks API Reference <https://docs.databricks.com/api/jobs/v2/run>
103. qandamaster338.xml <https://www.chegg.com/xml/qandamaster338.xml>
104. 30-Day Python Learning Roadmap | PDF <https://www.scribd.com/document/849701268/30-days-roadmap-learning-python>
105. Python Apps on Visual Studio Code <https://dokumen.pub/python-apps-on-visual-studio-code-develop-apps-and-utilize-the-true-potential-of-visual-studio-code.html>
106. Simple Perplexity AI clone https://colab.research.google.com/github/mklarqvist/ai-playground/blob/main/perplexity-clone/perplexity_clone.ipynb
107. qandamaster415.xml <https://www.chegg.com/xml/qandamaster415.xml>

108. 02-exploratory-data-analysis.ipynb <https://colab.research.google.com/github/glopasso/capstone/blob/master/extractive/notebooks/02-exploratory-data-analysis.ipynb>

109. bert_semantic_search_example.ipynb - Colab https://colab.research.google.com/drive/1P-kpzeG2dbc-DDxIhLpb_ibHF0jWg_Dn?usp=sharing